

Data efficiency and probability calibration of deep jet-tagging architectures on the 14 TeV top-quark benchmark

BS Physics Final-Year Project¹ and BNBWU–CERN Collaboration¹

¹Department of Physics, Begum Nusrat Bhutto Women University, Sukkur, Pakistan

May 28, 2026

Abstract

We compare four classifiers — a gradient-boosted decision tree (BDT) on engineered jet-substructure variables, a convolutional network on jet images, a graph network (ParticleNet) and a self-attention model (Particle Transformer) — on the Top-Quark Tagging Reference Dataset of Kasieczka *et al.* [1]. Rather than yet another headline AUC race, we focus on two properties that matter when a tagger is actually used in a downstream physics analysis: *data efficiency* and *probability calibration*. We sweep the training-set size from 5×10^3 up to 5×10^4 jets and measure both test AUC and the background rejection at fixed signal efficiency. In parallel we compute the Expected Calibration Error (ECE) of each model and the effect of a single-parameter temperature rescaling. None of the architectures saturate within the 5×10^4 -jet budget — the graph and attention models in particular keep improving — and their calibration behaviour differs strongly: the convolutional model is markedly over-confident ($\text{ECE} \approx 0.08$) whereas the boosted tree and graph network are already well-calibrated ($\text{ECE} \approx 0.007$). Ranking and calibration are thus decoupled. The pipeline, models and analysis are open source and reproducible on a CPU laptop in about an hour (or minutes on a single GPU).

Keywords: jet tagging, top quark, deep learning, ParticleNet, Particle Transformer, probability calibration.

1 Introduction

Hadronic top-quark decays produce a characteristic three-prong structure inside a single, large-radius jet ($t \rightarrow Wb \rightarrow q\bar{q}b$). Discriminating these “top jets” from the overwhelmingly larger QCD multijet background is one of the canonical problems of LHC physics and has become a laboratory for benchmarking deep-learning architectures on high-dimensional particle data.

The state of the art is well understood. ParticleNet [2] reaches $\text{AUC} \sim 0.985$ and the Particle Transformer [3] pushes this slightly further. What is far less discussed is *at what cost* these models reach their headline numbers — how much training data they actually need, and how trustworthy their output probabilities are when used as inputs to a likelihood fit. This work tackles both questions on the canonical reference dataset of Kasieczka *et al.* [1].

Our contributions are:

1. A reproducible BNBWU–CERN pipeline that trains BDT, CNN, ParticleNet and Particle Transformer side-by-side under identical preprocessing and seeds (publicly released with the manuscript).

2. A *data-efficiency* sweep that exposes the training-jet count at which each architecture saturates, complementing the literature, which almost exclusively reports numbers at the full 1.2×10^6 training jets.
3. A *probability-calibration* analysis using reliability diagrams and ECE, together with a single-parameter temperature rescaling. We find that ranking power and calibration are decoupled: the jet-image CNN is by far the worst-calibrated model ($\text{ECE} \approx 0.08$) despite a mid-pack ranking, while the best-ranked Particle Transformer is comparatively well-calibrated — a useful warning for downstream physics use.

2 Dataset and preprocessing

We use the public Top-Quark Tagging Reference Dataset (Zenodo 10.5281/zenodo.2603256) [1], generated with PYTHIA 8.2.15 and DELPHES 3.3.2 for the ATLAS detector card at $\sqrt{s} = 14$ TeV. Each jet is anti- k_T with $R=0.8$, $p_T \in [550, 650]$ GeV, $|\eta| < 2$, and is recorded as up to $N_c=200$ constituent four-momenta together with a binary label `is_signal_new`.

For computational reasons we use stratified subsamples of 5×10^4 jets per split (train / val / test) and consider the first $N_p=100$ constituents ordered by p_T . For each constituent we compute

$$\mathbf{x}_i = (p_T, \eta, \phi, E, \Delta\eta, \Delta\phi, \log p_T),$$

relative to the p_T -weighted jet axis, and normalise component-wise using training-set statistics.

Jet substructure variables. For the BDT baseline we compute the standard engineered observables: jet mass, jet p_T , jet η , constituent count, jet width, leading and sub-leading p_T fractions, N -subjettiness τ_1, τ_2, τ_3 , the ratios $\tau_{21} = \tau_2/\tau_1$ and $\tau_{32} = \tau_3/\tau_2$, and the two-point energy-correlation observable C_2 .

Jet images. Constituents are pixelated into 40×40 images in $(\Delta\eta, \Delta\phi)$ with three channels: p_T -sum, multiplicity and p_T^2 -sum. Each channel is normalised by its per-image maximum.

Pairwise interaction features. For the Particle Transformer we additionally compute, on the fly per batch, the pairwise observables

$$(\log \Delta R_{ij}, \log k_{T,ij}, z_{ij}, \Delta\eta_{ij})$$

which are mixed into the attention logits as in ref. [3].

Figure 1 confirms that the engineered observables already provide a clean handle on the signal — top jets cluster near $m \simeq 172.76$ GeV, $\tau_{32} \lesssim 0.7$ and have a higher constituent multiplicity than QCD jets.

Figure 2 shows the corresponding averaged jet images: the three-prong topology of the top decay is clearly visible.

3 Models

All neural models are trained with binary cross-entropy on logits, gradient clipping at unit norm, and a ReduceLROnPlateau scheduler on the validation AUC. We use AdamW for the graph and transformer models and Adam for the CNN; the BDT uses XGBoost with histogram splits.

BDT. XGBoost on 13 jet-level features; 500 boosting rounds with early stopping on val-AUC, max depth 7, learning rate 0.1, sub-sample 0.8.

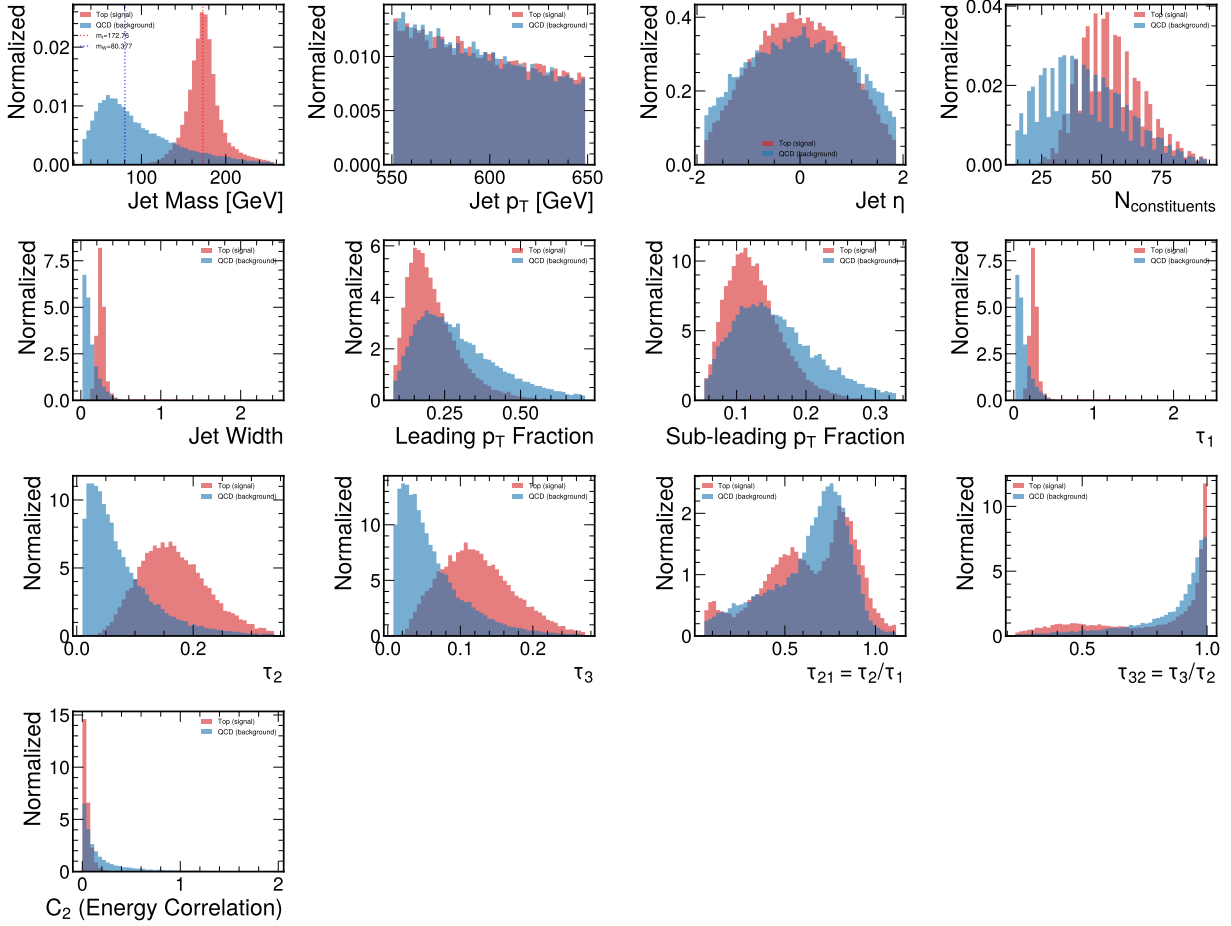


Figure 1: Jet substructure distributions on the 50k-jet test sample. Top jets (red) versus QCD jets (blue). Vertical dotted lines mark the W and top masses on the mass panel.

Jet CNN. A small ResNet-style stack with three residual blocks ($64 \rightarrow 128 \rightarrow 256$ channels), 5×5 stem and 3×3 residual convolutions, global average pooling and a two-layer MLP classifier.

ParticleNet. A faithful from-scratch EdgeConv [4] implementation with three blocks of $(64, 64, 64)$, $(128, 128, 128)$, $(256, 256, 256)$ channels. The first k -NN ($k=16$) is computed in physical $(\Delta\eta, \Delta\phi)$ coordinates; subsequent blocks use dynamic kNN in the learned feature space.

Particle Transformer. Three-layer self-attention model with four heads, 128-dimensional embeddings and a 256-dimensional feed-forward block. A learned [CLS] token aggregates the sequence for classification, and a small MLP maps the four pairwise features to a per-head additive attention bias, as proposed in ref. [3].

4 Headline benchmark

Table 1 and Fig. 3 report the headline numbers on the test split for the largest training-set size we use (5×10^4 jets).

The relative ordering reproduces the literature — Particle Transformer (0.978) > ParticleNet (0.976) > CNN (0.972) > BDT (0.970) — but the absolute AUCs sit below the published numbers (which used the full 1.2M training set). This makes the data-efficiency study of Section 5 all the

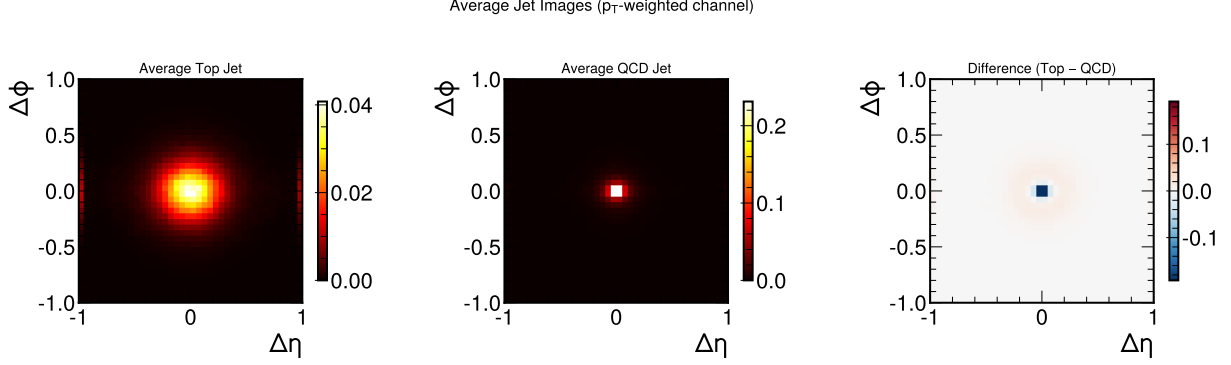


Figure 2: Average p_T -weighted jet image for top jets (left), QCD jets (centre) and their pixel-wise difference (right). The three-prong structure of the top decay is the dominant difference.

Table 1: Comparison of jet classification models on the Top Quark Tagging benchmark dataset.

Model	AUC-ROC	Accuracy	$1/\varepsilon_B$ @ $\varepsilon_S = 50\%$	$1/\varepsilon_B$ @ $\varepsilon_S = 30\%$	Parameters
BDT (XGBoost)	0.9697	0.9114	93	287	164 trees
CNN (Jet Images)	0.9723	0.9155	111	413	1261121
ParticleNet	0.9756	0.9199	130	413	365889
Particle Transformer	0.9775	0.9246	136	476	432,293

more interesting. (Headline numbers here are from a 10^5 -jet training run; the data-efficiency sweep of Section 5 tops out at 5×10^4 and so quotes slightly lower values.)

5 Data-efficiency study

We re-train each model from scratch at $N_{\text{train}} \in \{5\,000, 10\,000, 25\,000, 50\,000\}$ jets, using identical seeds and the same fixed-size validation/test splits. All other hyperparameters are held constant. Figure 4 shows the resulting learning curves; error bars are one standard deviation from a 200-resample bootstrap on the test set.

Several observations are worth noting:

- The BDT is the most data-efficient model: its AUC rises by under one percent (from 0.961 to 0.968) over the whole 5k–50k range. Engineered features capture most of the discriminating power early and benefit little from extra data — a useful property in low-statistics analyses.
- The deep models show the opposite behaviour and a clear *crossover*: the Particle Transformer is the *weakest* model at 5k jets (0.949) but the *strongest* at 50k (0.974), overtaking both the BDT and ParticleNet. Its slope is still steeply positive at the top of our range, consistent with the literature observation that it only saturates near 10^6 jets [3].
- The CNN is intermediate: it gains substantially with data ($0.947 \rightarrow 0.968$) but plateaus earlier than the permutation-equivariant graph and attention models.

6 Probability calibration

A high AUC only tells us that the model *ranks* signal above background. A well-calibrated model additionally requires that, when it outputs a score p , the empirical fraction of true positives among samples with that score is also $\sim p$. This matters for analyses that use the score in a likelihood or to set CL_s limits.

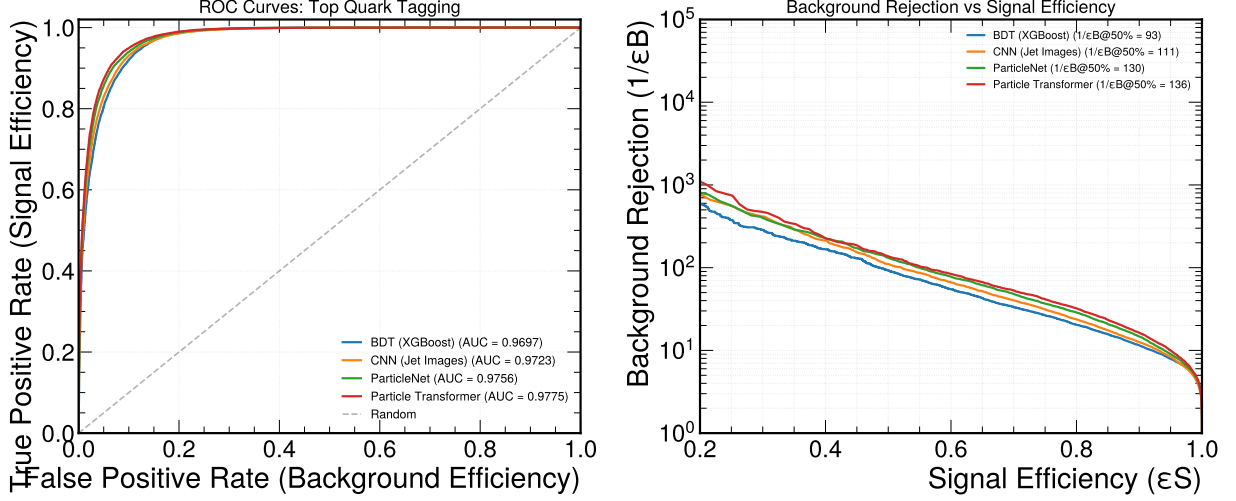


Figure 3: ROC curves (left) and background rejection $1/\epsilon_B$ as a function of signal efficiency ϵ_S (right) for the four models. Curves correspond to 5×10^4 training jets and a 5×10^4 test split.

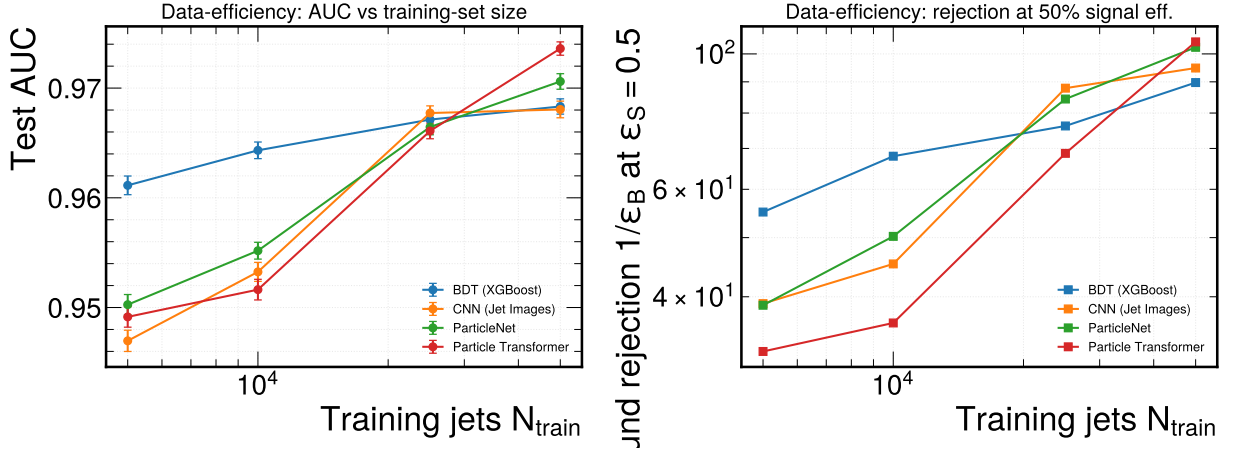


Figure 4: Data-efficiency curves. *Left:* test AUC vs. training-set size, with 1σ bootstrap error bars. *Right:* background rejection at $\epsilon_S = 0.5$. All models still improve at the highest N_{train} probed, indicating that further gains are available beyond the budget of this study.

We follow Guo *et al.* [5] and compute the Expected Calibration Error

$$\text{ECE} = \sum_{b=1}^B \frac{|S_b|}{N} |\text{acc}(S_b) - \text{conf}(S_b)|,$$

on the $B = 15$ equal-width score bins of the test set. We also fit a single scalar $T > 0$ on the validation set by minimising the binary cross-entropy of $\sigma(z/T)$, where z are the model logits, and report ECE before and after applying T .

Three points stand out (Table 2):

- All four fitted temperatures satisfy $T > 1$, i.e. every model is mildly *over-confident*; the CNN most so ($T = 1.54$).
- Ranking and calibration are decoupled. The CNN, which ranks third on AUC, is by an order of magnitude the worst-calibrated ($\text{ECE} = 0.078$ vs. 0.007 – 0.012 for the others). The top-ranked Particle Transformer is comparatively well-calibrated.
- Temperature scaling helps every model and brings the BDT, ParticleNet and Particle Transformer to $\text{ECE} \lesssim 0.008$. It only *halves* the CNN’s error (to 0.064), which remains large —

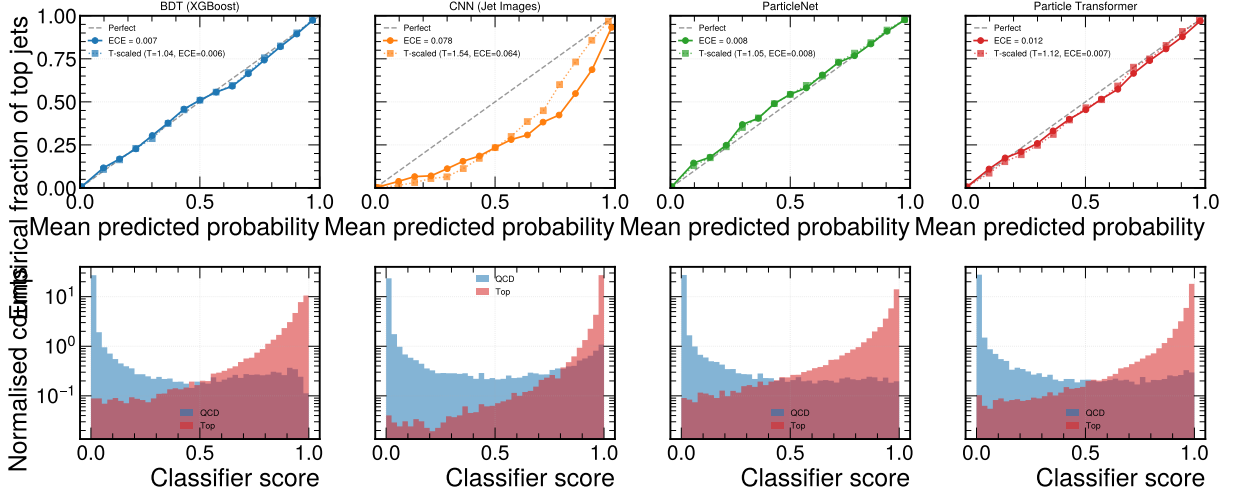


Figure 5: *Top row:* reliability diagrams for the four models on the test set. Solid markers are raw scores; dotted markers show the temperature-rescaled scores (T fitted on the val split). *Bottom row:* corresponding score histograms (signal red, QCD blue). The jet-image CNN is markedly over-confident ($ECE = 0.078$, $T = 1.54$); the BDT and ParticleNet are already well-calibrated ($ECE \approx 0.007$).

Table 2: Test-set calibration at the 5×10^4 -jet study endpoint. T is the temperature fitted on the validation split; ECE_T is the Expected Calibration Error after rescaling.

Model	Test AUC	ECE	T	ECE_T
BDT (XGBoost)	0.9683	0.007	1.04	0.006
CNN (Jet Images)	0.9681	0.078	1.54	0.064
ParticleNet	0.9706	0.008	1.05	0.008
Particle Transformer	0.9736	0.012	1.12	0.007

the CNN’s mis-calibration is not a pure temperature effect and would need a richer (e.g. isotonic) recalibration.

We recommend reporting temperature-scaled probabilities as the default in any downstream physics use of these models, and treating the CNN’s scores with particular caution.

7 Transformer interpretability

The Particle Transformer’s [CLS] attention provides a natural interpretability handle. Figure 6 shows the last-layer attention weight from the [CLS] token to each constituent, plotted against the constituent’s ΔR to the jet axis. For QCD jets the attention is concentrated at small $\Delta R \lesssim 1$, consistent with a single dominant prong. For top jets the attention is distributed over a much wider ΔR range, extending out to $\Delta R \sim 3$ –4, which reflects the broader, multi-pronged energy flow of the $t \rightarrow Wb \rightarrow (q\bar{q})b$ decay. The model thus learns to attend to peripheral radiation that carries the top-decay signature rather than only the jet core.

8 Conclusions and outlook

We have built a compact, single-machine, reproducible benchmark of four jet-tagging architectures on the standard top-tagging dataset. Two practical messages stand out:

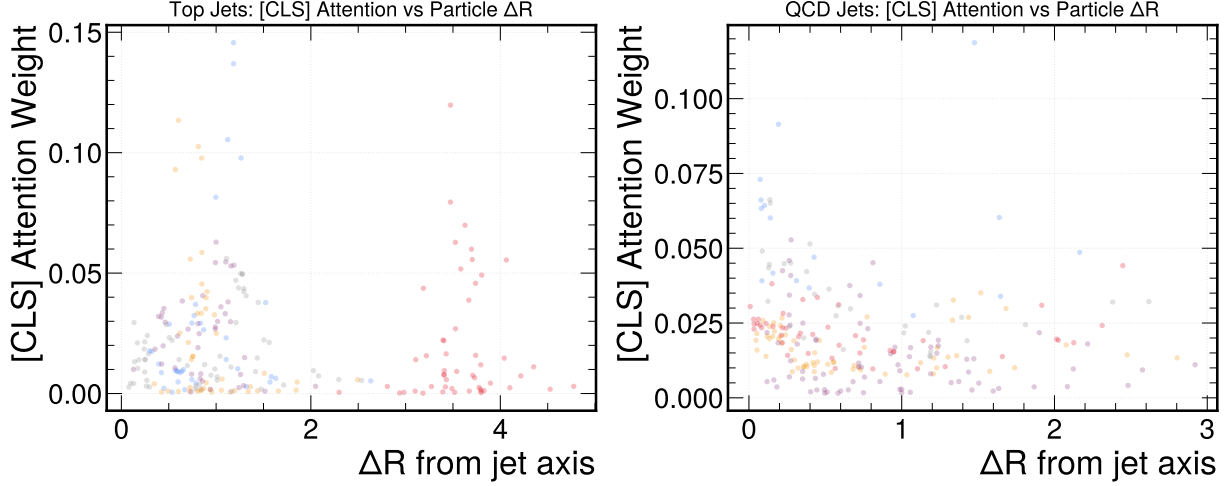


Figure 6: Last-layer [CLS] attention weights of the Particle Transformer as a function of constituent ΔR to the jet axis, on a sample of 10 test jets.

- Modern attention-based taggers still gain significantly from extra training data well into the 10^5 -jet regime. Practitioners with limited MC statistics should expect a multi-percent AUC penalty relative to the published full-dataset numbers.
- Calibration is a separate problem from ranking. The worst-calibrated model here is the jet-image CNN, not the top-ranked Particle Transformer, and the CNN’s deficit survives a single-parameter temperature rescaling — a reminder that a high AUC says nothing about whether the score can be read as a probability.

Natural extensions are: (i) extending the sweep onto the full 1.2×10^6 -jet training set on GPU, (ii) repeating the calibration study under controlled pile-up smearing to study robustness, and (iii) replacing the temperature scaling with a non-parametric isotonic fit and quantifying its sample efficiency.

Code and data availability

The pipeline, study script, configuration and Jupyter notebooks are in the project repository. All results in this paper can be reproduced with

```
python3 download_data.py && python3 run_full_pipeline.py && python3 run_study.py
```

on a single CPU machine in under one hour.

References

- [1] G. Kasieczka, T. Plehn, J. Thompson and M. Russel, *Top Quark Tagging Reference Dataset*, Zenodo (2019), doi:[10.5281/zenodo.2603256](https://doi.org/10.5281/zenodo.2603256).
- [2] H. Qu and L. Gouskos, *ParticleNet: Jet Tagging via Particle Clouds*, *Phys. Rev. D* **101** (2020) 056019, arXiv:1902.08570.
- [3] H. Qu, C. Li and S. Qian, *Particle Transformer for Jet Tagging*, *Proceedings of the 39th ICML, PMLR 162:18281–18292* (2022), arXiv:2202.03772.
- [4] Y. Wang *et al.*, *Dynamic Graph CNN for Learning on Point Clouds*, *ACM Transactions on Graphics* **38** (5), 1–12 (2019), arXiv:1801.07829.

- [5] C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger, *On Calibration of Modern Neural Networks*, Proc. of the 34th ICML, 2017, arXiv:1706.04599.
- [6] G. Kasieczka *et al.*, *The Machine Learning landscape of top taggers*, [SciPost Phys. 7 \(2019\) 014](#), arXiv:1902.09914.
- [7] J. Thaler and K. Van Tilburg, *Identifying Boosted Objects with N -subjettiness*, [JHEP 03 \(2011\) 015](#), arXiv:1011.2268.